

AI-BASED E-COMMERCE CUSTOMER CHURN PREDICTION AND RETENTION RECOMMENDATION SYSTEM

Artificial Intelligence & Data Science II — Mini Project Report

Self-contained prototype with synthetic demonstration data.

Abstract

Customer churn is a major business problem in e-commerce. This project predicts churn probability from behavioral features, classifies risk, and recommends a retention action. Random Forest is the primary model and Gradient Boosting is compared. A Streamlit dashboard provides interactive predictions. The bundled prototype uses synthetic data so it runs immediately.

1. Introduction

E-commerce behavioral signals can be transformed into predictive features for identifying customers who may stop purchasing. This project focuses on churn prediction and actionable retention recommendations rather than lifetime-value or survival-analysis modeling.

2. Problem Statement

Develop an AI-based system that predicts customer churn probability, classifies customers into risk levels, and recommends a practical retention action through an interactive web interface.

3. Objectives

- Clean customer behavioral data.
- Engineer behavioral features.
- Train and compare supervised classification models.
- Evaluate accuracy, precision, recall, F1 and ROC-AUC.
- Generate risk categories and retention recommendations.
- Provide a Streamlit interface.

4. Methodology

Customer Data → Preprocessing → Behavioral Features → Train/Test Split → Random Forest and Gradient Boosting → Churn Probability → Risk Category → Retention Recommendation → Streamlit Dashboard.

The demonstration dataset contains 1,500 synthetic customer profiles. Numeric variables are standardized in a scikit-learn pipeline and a stratified 75/25 split is used.

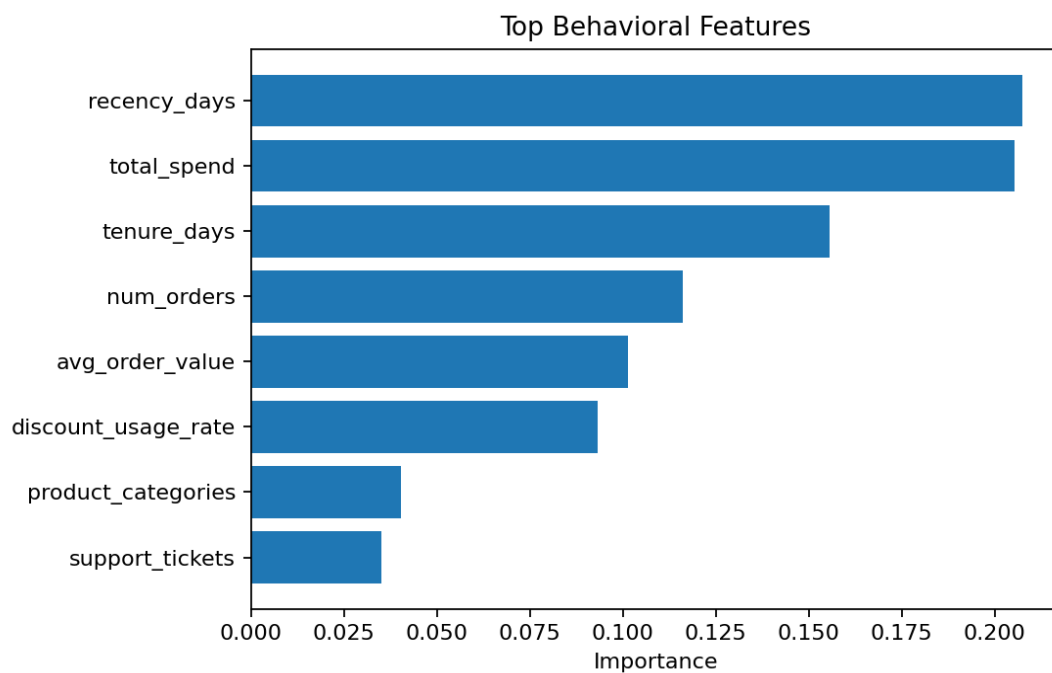
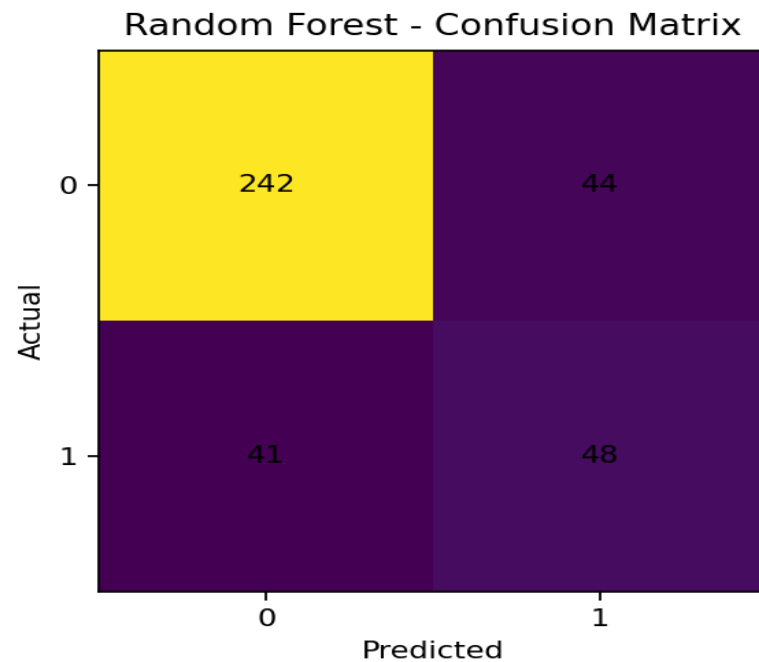
5. Technology Stack

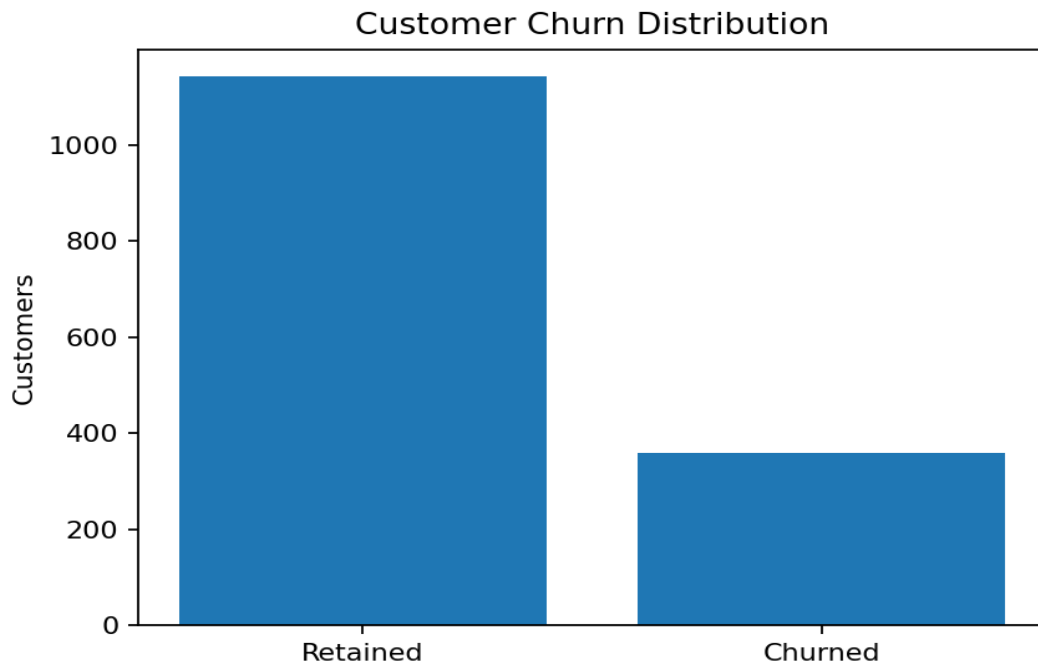
Component	Technology
Language	Python
Processing	Pandas, NumPy
ML	scikit-learn
Models	Random Forest, Gradient Boosting
UI	Streamlit
Visualization	Matplotlib
Model storage	joblib

6. Results

Selected model: Random Forest.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.773	0.522	0.539	0.530	0.787
Gradient Boosting	0.787	0.569	0.416	0.481	0.775





7. Streamlit Application

The interface accepts customer profile attributes and displays churn probability, risk level, behavioral features and a retention recommendation. Launch with ``streamlit run app.py``.

8. Limitations

The bundled data is synthetic. Therefore, metrics are prototype results and should not be presented as production performance. Real deployment requires an approved transaction dataset, a documented churn definition and temporal validation.

9. Conclusion

The project demonstrates a complete AI workflow from customer behavior to predictive churn risk and actionable retention guidance. It is compact enough for a mini project and can be extended with real data and explainable AI.

10. References

- Breiman, L. Random Forests. Machine Learning, 45, 5–32, 2001.
- Friedman, J. H. Greedy Function Approximation: A Gradient Boosting Machine. Annals of Statistics, 2001.
- Scikit-learn documentation.
- Streamlit documentation.